Momentum analysis: put it together

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use a model to compare a deflection option and its limits.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 10.*
5 m/s west and 7 m/s south.

02. Worked example: explain the model
Internal changes cancel; total momentum change zero under negligible external impulse.

03. Read and interpret the model
Δv=J/M uses mass; shift=Δv t uses time. Units N·s, m/s, m.

04. Apply the physics
Δv=0.1 m/s; impulse 10 N·s.

05. Explain your reasoning
Actual Δv would be smaller because mass is larger; shift smaller at same time in model.

06. Review: connect an earlier idea *Review: Unit 3, Day 1.*
Change magnitude 24 kg·m/s; force 8 N opposite initial motion.

07. Exit check
10 m; constant added transverse velocity/straight-line comparison.

08. STRETCH: test the model
Final total 9 kg·m/s; final speed 3 m/s right.

Momentum analysis: put it together

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Use a model to compare a deflection option and its limits.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 10.*
Other p=(12, -5) kg·m/s; magnitude≈13 kg·m/s.

02. Worked example: explain the model
Internal changes cancel; total momentum change zero under negligible external impulse.

03. Read and interpret the model
Δv=J/M uses mass; shift=Δv t uses time. Units N·s, m/s, m.

04. Apply the physics
Need .005 m/s and 1×10^{7} N·s. Option gives .004 m/s, shift 16 km; short by 4 km in model.

05. Explain your reasoning
Actual Δv would be smaller because mass is larger; shift smaller at same time in model.

06. Review: connect an earlier idea *Review: Unit 3, Day 1.*
J=-24 N·s; $F_{avg}\approx$-8 N, left.

07. Exit check
Unchanged: shift=(J/M) t; M×2 halves Δv, t×2 restores shift.

08. STRETCH: test the model
Initial total 6; final 3; v=1 m/s. External impulse changes system momentum.